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Self-learning style transfer via dual-path feature fusion in diffusion models

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ABSTRACT

Style transfer combines the content of one image with the artistic style of another. Recent advances in diffusion models have improved image generation, but reducing style-content coupling while maintaining high-quality generation remains challenging. We present Self-Learning Style Transfer (SLST), a diffusion-based framework that coordinates style-dominant and content-oriented representations through dual-path feature fusion. For style representation, the Self-Learning Style Encoder (SLS-Encoder) uses learnable queries, random masking, and cross-view consistency learning to encourage style-dominant features with reduced content dependence, without requiring paired content-style data. For content preservation, adaptive edge extraction provides structural cues, while BLIP-2 generates concise content-focused descriptions for semantic guidance. Quantitative evaluations, visual comparisons, and user studies indicate favorable style-transfer behavior in the evaluated settings. These downstream results suggest reduced content interference but do not directly establish representation-level disentanglement.

1. Introduction

Artistic images are widely recognized as powerful tools for communication, capable of conveying profound thoughts and emotions through the strategic application of color, composition, and texture. These elements shape the visual experience of the viewer and can elicit emotional responses. In recent years, style transfer techniques have emerged as a way to connect artistic expression with everyday applications, enabling images that combine different artistic styles with real-world content. These techniques make artistic expression more accessible to users with different levels of artistic training.

Early style transfer approaches adopted a CNN-based feature separation and recombination paradigm, focusing on statistical learning of style from reference images. Notably, Li and Wand (2016) laid the foundation by combining texture synthesis and neural networks, while Gatys et al. (2016) introduced a method based on convolutional neural networks to separate and recombine content and style representations, albeit with relatively high computational cost. Huang and Belongie (2017) achieved real-time arbitrary style transfer, but relied on low-level features such as color and texture and could not always capture fine-grained style differences. The advent of GANs (Radford, 2015) introduced a GAN-based adversarial training paradigm, shifting from feature matching to end-to-end style distribution learning. CycleGAN (Zhu et al., 2017) enabled unpaired transfer via cycle-consistency losses, overcoming the need for paired datasets but facing mode collapse and training instability. StyleGAN (Karras et al., 2019) achieved hierarchical style control through progressive growing, yet could struggle to reproduce complex artistic styles.

Recent advances in generative models, particularly diffusion models (Chen et al., 2023b; Xia et al., 2023; Zhang et al., 2023b), have expanded the design space for style transfer. Diffusion models possess remarkable generative capabilities, allowing them to produce highly realistic and stylized images conditioned on textual prompts. Text-based style transfer prompts show promising results (Avrahami et al., 2022; Chai et al., 2023; Chen et al., 2023a; Sohn et al., 2024), where predefined labels, such as “cartoon”, “sketch”, or “watercolor”, guide the image generation process. However, this approach is constrained by the availability of labeled datasets that adequately represent diverse artistic styles. Many complex artistic styles are difficult to describe with simple words, which cannot fully capture the richness of an individual artistic expression. As a result, the use of images as direct stylistic references has become a more compelling direction, with images inherently carrying fine-grained, complex stylistic details that text cannot fully convey. However, a key challenge in image-based style transfer is to learn style-dominant representations with reduced dependence on the semantic content of the style reference. Existing methods have obvious shortcomings. LoRA (Hu et al., 2021), TextInversion (Gal et al., 2022), and DreamBooth (Ruiz et al., 2023) capture a target style by fine-tuning

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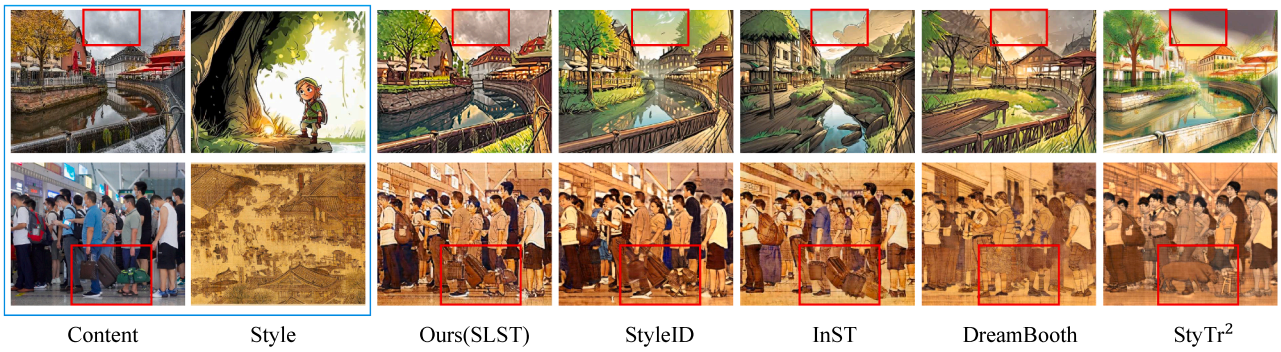


Fig. 1. Visual comparison of the proposed SLST and state-of-the-art methods. The red boxes highlight local regions in complex scenes. The first row compares the shape and color of dark clouds, and the second row compares the position and shape of a suitcase. The examples illustrate differences among the methods in style transfer and content preservation.

the diffusion model, but this approach is resource-intensive and may be impractical for real-world applications. While DEADiff (Qi et al., 2024) attempts separate extraction of style and content via a trained Q-former, and IP (Ye et al., 2023) adopts CLIP (Radford et al., 2021) as the image encoder for style extraction, the text-image paired training mechanism of CLIP can encourage semantic content and style to co-occur in the extracted features, which may limit the separation of style and content information.

To reduce style-content coupling in image-based style transfer, we propose Self-Learning Style Transfer (SLST), a diffusion-based framework that coordinates style-dominant representation learning with complementary conditioning paths (Garcia-Dominguez et al., 2024; Zhang et al., 2025b).

As illustrated in Fig. 2, SLST takes content and style images as inputs and combines style representation learning with complementary feature-fusion paths. For style images, the Self-Learning Style Encoder (SLS-Encoder) uses random masking to create multiple views for self-supervised training and learns fine-grained style-dominant features with reduced content dependence. For content images, adaptive Canny edge detection extracts structural cues, while BLIP-2 generates concise content-focused descriptions that provide semantic guidance with reduced style interference.

During inference, the complementary features are fused through separate conditioning paths. The content edge features and style embeddings extracted by the SLS-Encoder are concatenated and supplied to ControlNet (Zhang et al., 2023a) to provide structural guidance. The content text features and style features are injected into separate cross-attention branches of the diffusion model. This design is intended to balance content structure and semantics with stylistic fidelity while helping reduce interference between the conditioning signals.

On top of this representation, SLST coordinates semantic text guidance, style guidance, and structural residual conditioning through complementary conditioning paths. Therefore, the contribution should be understood as a new style-representation learning strategy together with its coordinated conditioning design, rather than as a new diffusion backbone.

In conclusion, our primary contributions are as follows.

- We propose the Self-Learning Style Transfer (SLST) framework for diffusion-based style transfer to reduce style-content coupling. The dual-path feature-fusion design injects style and text through separate cross-attention branches and uses structural cues through a separate ControlNet path, helping reduce interference between guidance signals while balancing stylistic fidelity and content preservation.
- The key Self-Learning Style Encoder (SLS-Encoder) is designed to learn style-dominant features from style images without requiring paired content-style data or additional annotations. Random masking creates multiple views of each style image, and a self-learning objective encourages consistency between masked-view embeddings and the corresponding unmasked reference embedding. This strategy is intended to reduce content dependence rather than eliminate all semantic information.
- The reported quantitative evaluations, visual comparisons, cross-architecture experiments, component ablations, and user-study results characterize SLST's style-transfer behavior in the evaluated settings. The results indicate a favorable empirical balance between stylistic consistency and content preservation, without constituting direct representation-level evidence of complete style-content disentanglement.

2. Related work

2.1. Diffusion model

Diffusion models have become an important class of generative models, offering strong image-generation quality and diversity compared with traditional approaches such as Generative Adversarial Networks (GANs) (Radford, 2015). These models operate by progressively denoising random noise to produce high-quality images, making them highly versatile for various tasks in computer vision.

One early work that integrated text features into diffusion models was GLIDE (Nichol et al., 2021), which used an attention mechanism to incorporate textual guidance during the denoising process. This approach improved controllability by enabling closer alignment between textual prompts and generated outputs. Building on this foundation, latent diffusion models (LDMs) (Rombach et al., 2022) introduce a Variational Auto-Encoder (VAE) (Kingma, 2013) to map the denoising process from pixel space to a low-dimensional latent space. This approach not only reduces computational costs but also improves scalability, allowing diffusion models to handle complex tasks more efficiently. Stable Diffusion represents an efficient implementation of LDM, and its open-source nature facilitates the development of many outstanding projects within the community (Chen et al., 2023b; Xia et al., 2023; Xu et al., 2024; Zhang et al., 2024). Extensions of diffusion models further expand their capabilities by integrating non-textual conditions. For example, ControlNet (Zhang et al., 2023a) enables fine-grained control over the generation process by incorporating external signals such as Canny edges, semantic layouts, and human poses. Similarly, IP-Adapter (Ye et al., 2023) leverages intermediate representations to enhance the model's ability to condition on diverse inputs, thereby improving both controllability and diversity.

2.2. Style transfer

Style transfer has long been a focal point in computer vision, aiming to seamlessly blend the content of one image with the artistic style of

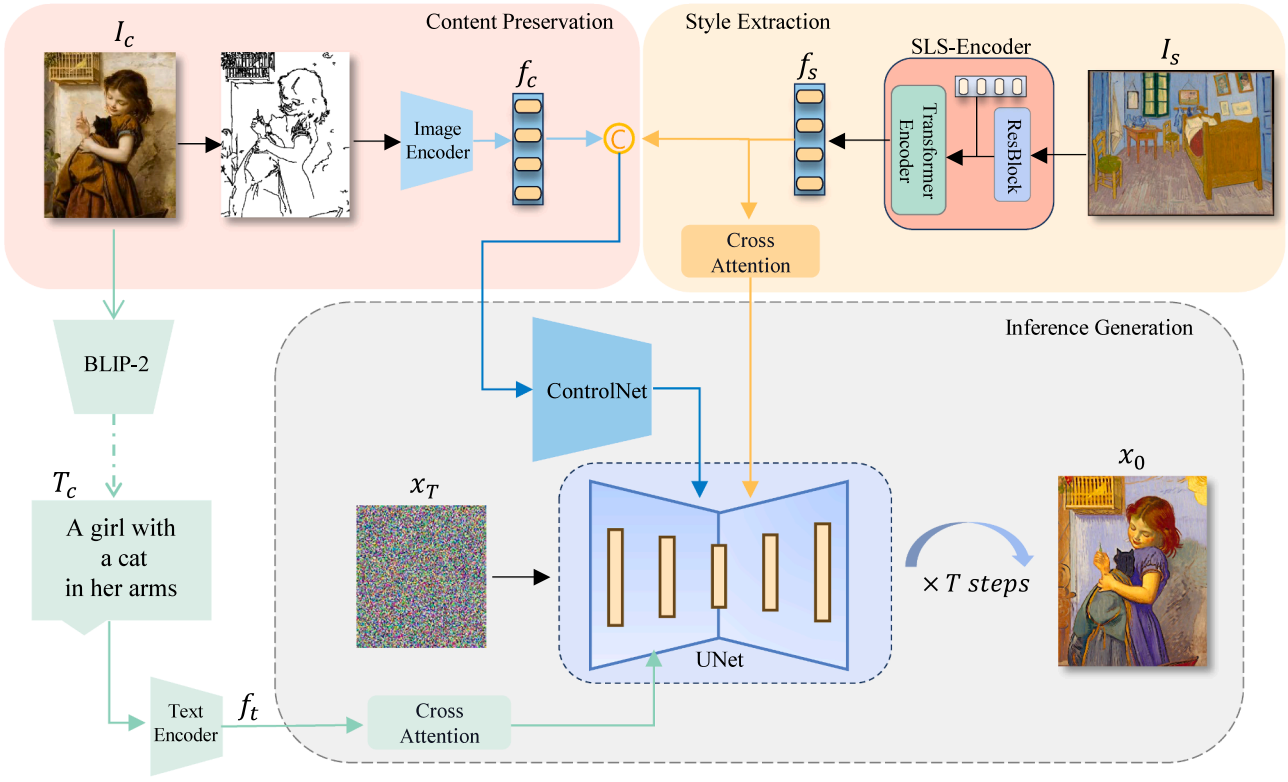


Fig. 2. The overall framework for Self-Learning Style Transfer. (a) Content Preservation Module. Edge features f_c provide structural cues from the content image, retaining major contours while reducing appearance-related information. BLIP-2 generates a concise content description T_c, which is used as text conditioning f_t. (b) Style Extraction Module. The SLS-Encoder extracts style embeddings $f_s \in \mathbb{R}^d$ intended to represent style-dominant attributes with reduced content dependence. (c) Inference Generation Module. The framework combines style embeddings f_s, text embeddings f_t, and edge embeddings f_c. Text and style are injected through separate cross-attention branches, while the edge and style features are combined as structural conditioning for ControlNet. These complementary paths are designed to balance stylistic appearance and content structure. After T denoising steps, the latent noise x_T is transformed into the final stylized image x_0.

another. Early methods primarily rely on handcrafted features or statistical learning to achieve this goal. Gatys et al. (2016) make use of neural networks for style transfer by employing the Gram matrix to represent style information extracted from convolutional layers. This method enables the separation of content and style through feature variance, laying the groundwork for subsequent research. AdaIN (Huang & Belongie, 2017) (Adaptive Instance Normalization) further advances the field by applying second-order statistics to normalize feature maps, facilitating real-time style transfer without requiring large datasets for training. Zhu et al. (2017) extend this work by introducing CycleGAN, a generative adversarial network-based approach that learns mappings between different domains, thus enabling unpaired image-to-image translation.

More recent developments incorporate Transformer (Vaswani, 2017) architectures to address the limitations of CNN-based methods in handling long-range dependencies and preserving fine details. StyleFormer (Wu et al., 2021) and StyTr^2 (Deng et al., 2022) exemplify this trend, leveraging Transformers to capture global context and enhance the expressiveness of style representation, consistent with the relational feature modeling paradigm proposed in Shao et al. (2025), which highlights the value of fine-grained relational learning for visual tasks. These approaches demonstrate superior performance in generating visually coherent and artistically rich images.

In parallel, diffusion models are increasingly explored for style transfer tasks. InST (Zhang et al., 2023b) utilizes CLIP (Radford et al., 2021) as a guiding mechanism to align the diffusion process with the target styles, achieving impressive results in the transfer of cross-modal styles. ArtAdapter (Chen et al., 2024b) introduces a multi-level style encoder to extract hierarchical style information, enabling finer control over the stylization process. DEADiff (Qi et al., 2024) employs a two-stage training strategy with Q-former (Li et al., 2023) to decouple style and content embeddings, offering greater flexibility in controlling the balance between these components. Notably, the cross-modality fusion idea from Zhang et al. (2025a) further supports the potential of integrating multi-source information for style transfer, while part-whole relational fusion (Liu et al., 2025) provides insights into effective decoupling of visual components, both of which inform our framework design. StyleShot (Gao et al., 2024) develops a Style-Aware Encoder grounded in diffusion model, focusing on capturing stylistic nuances while preserving content fidelity. Despite these advances, existing methods often struggle to fully decouple style and content information, leading to residual content leakage (Wu et al., 2019) in style representation. Additionally, capturing intricate style details remains challenging due to the inherent complexity of artistic expressions.

3. Method

To reduce style-content coupling in style transfer, this paper proposes a Self-Learning Style Transfer (SLST) framework grounded in the Stable Diffusion architecture. The framework encompasses three modules: a self-learning style encoder (SLS-Encoder), a content preservation module, and an inference generation module. The SLS-Encoder learns style-dominant features without paired content-style training data; the content preservation module provides structural and semantic cues from content images; and the inference generation module combines these complementary conditions to balance stylistic consistency and content fidelity. The following sections describe the technical foundation and module design.

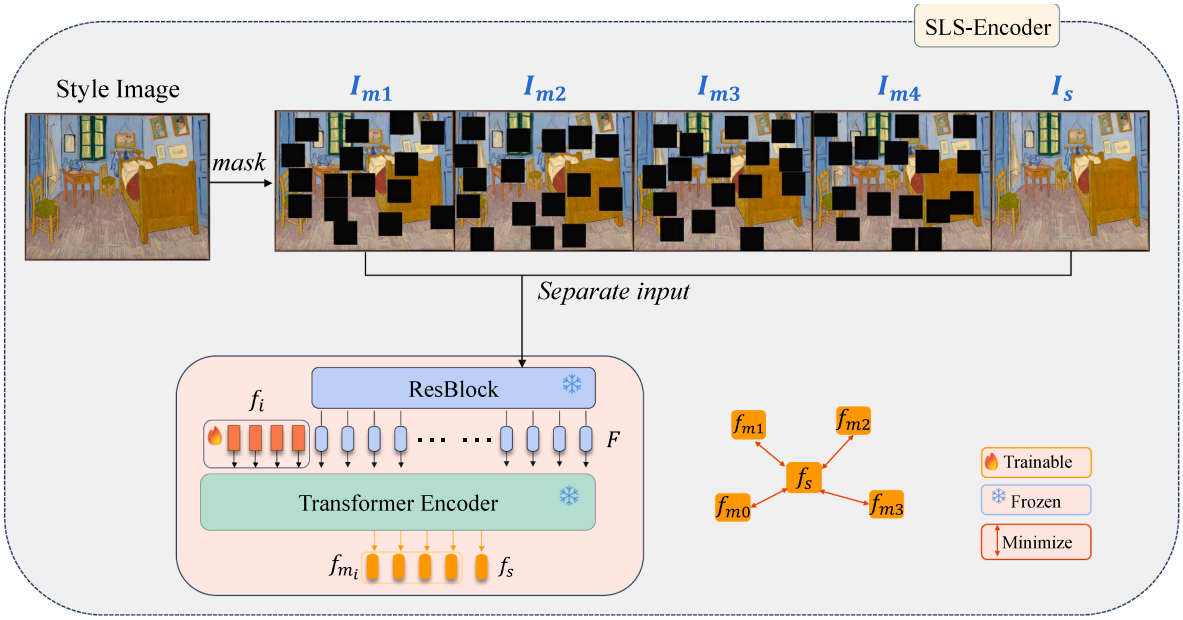


Fig. 3. The training process for SLS-Encoder. Random masks partially disrupt spatial and object-level semantic cues while leaving local stylistic statistics observable in unmasked regions. The ResBlock and Transformer Encoder process the masked views and the original style image, while learnable queries aggregate image-conditioned features. The consistency and mean/variance-alignment objectives encourage masked-view embeddings to remain consistent with the corresponding unmasked reference. This objective reduces sensitivity to mask-specific cues but does not guarantee complete removal of semantic information.

3.1. Preliminary on stable diffusion

Latent Diffusion Model (LDM) (Rombach et al., 2022) is a diffusion model trained in a low-dimensional latent space, which reduces computational costs while enhancing the understanding of semantic information. The given image $x \in \mathbb{R}^{H\times W\times 3}$ is encoded by the VAE encoder E into a latent representation $z_0 \in \mathbb{R}^{h\times w\times C_z}$, where $C_z = 4$ for Stable Diffusion v1.5. During the forward diffusion process, Gaussian noise is added according to the noise schedule, yielding $z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$. At the final timestep, z_T approximately follows a standard Gaussian distribution. The reverse process gradually denoises the sequence $z_T, z_{T-1}, \ldots, z_1, z_0$ to recover the target image. The standard simplified noise-prediction objective for the U-Net, $\epsilon_\theta(z_t, t, c)$, is defined as follows:

$$\mathcal{L} = \mathbb{E}_{z_0,c,\epsilon\sim\mathcal{N}(0,1),t}\left[\left\|\epsilon - \epsilon_\theta\left(z_t, t, c\right)\right\|^2\right] \tag{1}$$

where $z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$, $\bar{\alpha}_t$ is the cumulative product of the noise schedule, t is the diffusion timestep, and c denotes additional conditioning information.

3.2. Self-learning style encoder

Due to the complexity and diversity of style images, reducing style-content coupling remains challenging. The lack of paired content-style datasets further complicates the learning of style-dominant representations. Inspired by DEADiff (Qi et al., 2024), we propose a self-learning style feature extraction module trained using style images without paired content images. Its objective is to encourage style-dominant representations with reduced content dependence, as illustrated in Fig. 3. Traditional style-transfer methods often rely on paired style-content data for supervised learning, but acquiring large-scale aligned data can be expensive and many artistic styles have limited training data.

To address this, this paper proposes a self-learning paradigm whose core idea is to automatically generate training pairs. This design aligns with the semi-supervised learning philosophy in Chen et al. (2024a), which proves that effective feature learning can be achieved with extremely limited labels by constructing pseudo-samples, overcoming the data scarcity bottleneck in traditional style transfer.

The model only requires style images for training, without additional paired content or annotations. Pseudo-training pairs are constructed by applying random masks to images, generating multi-view samples. Through statistical alignment and comparison losses, the model is encouraged to reduce sensitivity to view-specific semantic cues and emphasize cross-view consistent stylistic characteristics, thereby reducing content leakage from style images. The mask generation uses non-overlapping rectangular blocks to preserve the integrity of local style features. Specifically, given a style image I_s, we randomly generate multiple mask areas (covering about 40% of the pixels) to generate a set of masked images I_{m1}, I_{m2}, I_{m3}, I_{m4}. Random masking is not assumed to completely remove semantic content. Instead, it acts as an inductive bias that disrupts spatial continuity, part configuration, and long-range object relationships differently across independently masked views, while local chromatic statistics, stroke texture, contrast, and other stylistic cues remain observable in the unmasked regions. By encouraging multiple masked views of the same reference image to produce consistent embeddings, the learning objective suppresses mask-sensitive information and favors view-consistent stylistic characteristics. Mean-alignment and variance-alignment further preserve statistical properties that remain stable across views. Nevertheless, global shapes, repeated motifs, iconic objects, or semantic cues strongly correlated with a particular style may survive masking. Therefore, the proposed strategy is intended to reduce, rather than eliminate, content dependence.

The original style image and its masked views first pass through a residual block (ResBlock), which is intended to preserve low-level visual details while extracting base features F. These features are combined with learnable query embeddings f_i, which aggregate image-conditioned information from the visible regions. The fused features are processed by the Transformer Encoder (Vaswani, 2017) to obtain embeddings f_{mi} for the masked views and f_s for the original style image. The query embeddings are optimized during training, while the pre-trained ResBlock and Transformer Encoder parameters are kept frozen. The encoder weights are initialized from the Style-Aware Encoder provided by StyleShot (Gao et al., 2024). Random masking disrupts spatial continuity and object-level relationships in different ways across views, encouraging the model to retain visual cues that remain stable

across masking patterns. The consistency objective uses the embedding of the corresponding unmasked style image as an image-specific reference rather than forcing all style images toward a shared target. It encourages masked-view embeddings to approach that reference, but does not guarantee that the embeddings contain style information exclusively. To quantify the difference between the embeddings of masked views and the original style image, we compute their cosine distance, defined as follows:

$$\mathrm{dist}_{\cos}\left(f_{mi}, f_s\right) = 1 - \frac{f_{mi} \cdot f_s}{\|f_{mi}\|\|f_s\|} \tag{2}$$

where $\cdot$ denotes the dot product and $\|\cdot\|$ denotes the L_2 norm. A smaller cosine distance indicates that a masked-view embedding is closer to the corresponding unmasked reference embedding. The mask loss can be obtained as:

$$\mathcal{L}_{fc} = \sum_{i=1}^{4} \left(\mathrm{dist}_{\cos}(f_{mi}, f_s)\right)^2 \tag{3}$$

The essence of style can be understood as the statistical regularities (Huang & Belongie, 2017) of certain features in an image. For instance, Impressionist styles exhibit high local continuity in color distribution with random brushstroke directions. Ink paintings feature strong light-dark contrast, marked by large brightness variance, low color saturation, and color means tending toward gray tones. Cartoon styles have sharp edges with high gradient variance, uniform color blocks, and concentrated color means. These statistical properties of different styles can be quantified through key metrics, and mean and variance stand out as effective tools for this purpose. The mean μ captures the central tendency of features, such as the dominant hue or average brightness. The variance σ^2 describes the range of variation in features, such as color diversity or textural complexity.

We introduce mean and variance alignment to address potential issues in masked image training. Mean alignment ensures the masked images maintain consistency with the original style in main tone and stroke density, avoiding style offset caused by local masking. Variance alignment constrains the diversity of style features, suppressing statistical noise introduced by mask areas.

$$\mathcal{L}_{\text{mean}} = \sum_{i=1}^{4} \left\|\mu\left(f_{mi}\right) - \mu\left(f_s\right)\right\|_2^2 \tag{4}$$

$$\mathcal{L}_{\text{var}} = \sum_{i=1}^{4} \left\|\sigma^2\left(f_{mi}\right) - \sigma^2\left(f_s\right)\right\|_2^2 \tag{5}$$

To encourage cross-view consistency in the style-dominant representations, the total training loss is defined as:

$$\mathcal{L} = \mathcal{L}_{fc} + \lambda_\mu \mathcal{L}_{\text{mean}} + \lambda_\sigma \mathcal{L}_{\text{var}} \tag{6}$$

Among them, λ_μ and λ_σ are equilibrium weights (default $\lambda_\mu = 1.0$, $\lambda_\sigma = 0.5$), respectively controlling the intensity of mean alignment and variance alignment.

The self-learning strategy based on random masking and mean/variance alignment encourages style-dominant representations with reduced content dependence. It is intended to reduce content interference, not eliminate it, and provides a basis for the downstream style-transfer experiments.

3.3. The architecture of self-learning style transfer (SLST)

The SLST framework is composed of three key modules: a content preservation module, a style extraction module, and an inference generation module. As shown in Fig. 2, these modules combine complementary conditions to improve content preservation and stylistic consistency.

In the content preservation module, it seems plausible for some methods (Ye et al., 2023) to directly use CLIP (Radford et al., 2021) to extract image content features. However, this approach (Cheng et al., 2019) may lead to the issue of content image style leakage. We use an adaptive Canny edge detector (Canny, 1986) to process content images, dynamically adjusting the thresholds to preserve critical structures (such as hair, textures) and avoid excessive smoothing or noise interference.

To obtain a content-oriented representation with reduced style interference, we process the content image I_c into an edge image. This step suppresses a substantial portion of appearance-related information while retaining content-relevant structural cues, which can help preserve the original layout and contours. BLIP-2 generates a concise content description T_c to provide semantic guidance and reduce semantic deviations during stylization. The SLS-Encoder is designed to learn style-dominant features $f_s \in \mathbb{R}^d$ with reduced dependence on content semantics.

In the inference generation module of the diffusion model, the concatenation and fusion of edge image content embeddings with style embeddings enable a harmonious integration of style and layout objects in the generated results. The edge image only contains low-frequency structure information, lacks high-frequency details, and the style embedding supplements the injection of high-frequency features to make up for the shortcomings of the edge image. After feature fusion, the complementary content and style conditions encourage the diffusion model to preserve the major structural layout while introducing appearance changes associated with the reference style. Additionally, ControlNet demonstrates powerful spatial layout control capabilities in diffusion model, where we input complementary edge image features and style features into ControlNet as residual control information. The output of ControlNet is the input to the U-Net upsampling process (Ronneberger et al., 2015).

Style and text embeddings represent different modalities. A conventional cross-attention layer is typically trained with text embeddings as conditioning, which may affect how style information is incorporated into the generated result. We use a decoupled cross-attention mechanism (Ye et al., 2023) to process text and style embeddings through parallel attention branches with independent key-value projections. Text embeddings provide semantic guidance, such as object categories and spatial relationships, while style embeddings guide appearance attributes, such as brushstrokes and color. This architecture separates the conditioning operations and is intended to help reduce interference; it does not by itself establish statistical independence between the learned representations. The calculation process is detailed as follows:

$$Z_{cond} = softmax(\frac{QK^t}{\sqrt{d}})V^t + \lambda \cdot softmax(\frac{QK^s}{\sqrt{d}})V^s \tag{7}$$

where Z denotes the latent feature of the U-Net at a given denoising step; $Q = ZW_q$ is the query shared by the two attention terms; $K_t = c_t W_k^t$, $V_t = c_t W_v^t$ and $K_s = c_s W_k^s$, $V_s = c_s W_v^s$ are the key and value matrices for the text and style branches respectively, with key-value projections instantiated independently for each branch; c_t serves as the text condition for content description and $c_s = f_s$ is the style embedding; and λ balances the two terms.

In each downsampling and upsampling layer of U-Net, the style coded f_s regulates the feature distribution through adaptive layer normalization:

$$\mathrm{AdaNorm}(x) = \gamma_s \cdot \frac{x - \mu(x)}{\sigma(x)} + \beta_s \tag{8}$$

where γ_s and β_s are the scale and shift parameters generated from the style embedding f_s through an MLP. $\mu(x)$ and $\sigma(x)$ denote the mean and standard deviation of the feature map x.

4. Experiments

4.1. Experiment settings

Training. We use pretrained IP-Adapter (Ye et al., 2023) and Stable Diffusion v1.5 parameters as the initialization for the frame-

Table 1
Quantitative results and user-study selection counts for the compared methods. Bold and underlined values indicate the highest and second-highest reported values, respectively, according to the direction of each metric. The first ten rows are existing methods, and the final two rows report SLST variants.

Method	Published in	Style Align		Content Align		Quality		User-Study	
		CLIP-S↑	FID↓	SSIM↑	CFSD↓	ImageReward↑	Aesth↑	User-SS↑	User-CS↑
AdaIN (Huang & Belongie, 2017)	ICCV 2017	0.315	69.255	0.259	0.316	−1.597	4.633	1	0
StyleGAN (Karras et al., 2019)	CVPR 2019	0.360	55.941	0.252	0.332	−1.488	4.618	0	1
ArtFlow (An et al., 2021)	CVPR 2021	0.472	39.588	0.341	0.307	−0.980	4.904	2	1
AdaAttN (Liu et al., 2021)	ICCV 2021	0.457	36.083	0.380	0.286	−1.103	5.120	1	2
StyTr^2 (Deng et al., 2022)	CVPR 2022	0.550	20.653	0.486	0.300	−0.496	5.295	3	4
DreamBooth (Ruiz et al., 2023)	CVPR 2023	0.516	23.387	0.418	0.289	0.384	5.378	2	4
InST (Zhang et al., 2023b)	CVPR 2023	0.538	21.311	0.507	0.676	0.168	5.251	7	5
Artbank (Zhang et al., 2024)	AAAI 2024	0.508	20.493	0.464	0.293	0.216	5.400	3	2
StyleID (Chung et al., 2024)	CVPR 2024	0.569	**18.831**	0.398	0.251	0.267	5.405	7	5
AD (Zhou et al., 2025)	CVPR 2025	0.531	21.368	0.494	0.275	0.089	5.342	5	3
SLST (+CLIP)	Ours	0.563	20.029	0.489	0.265	**0.396**	5.408	6	8
SLST	Ours	**0.590**	18.994	**0.543**	**0.228**	0.392	**5.411**	**11**	**13**

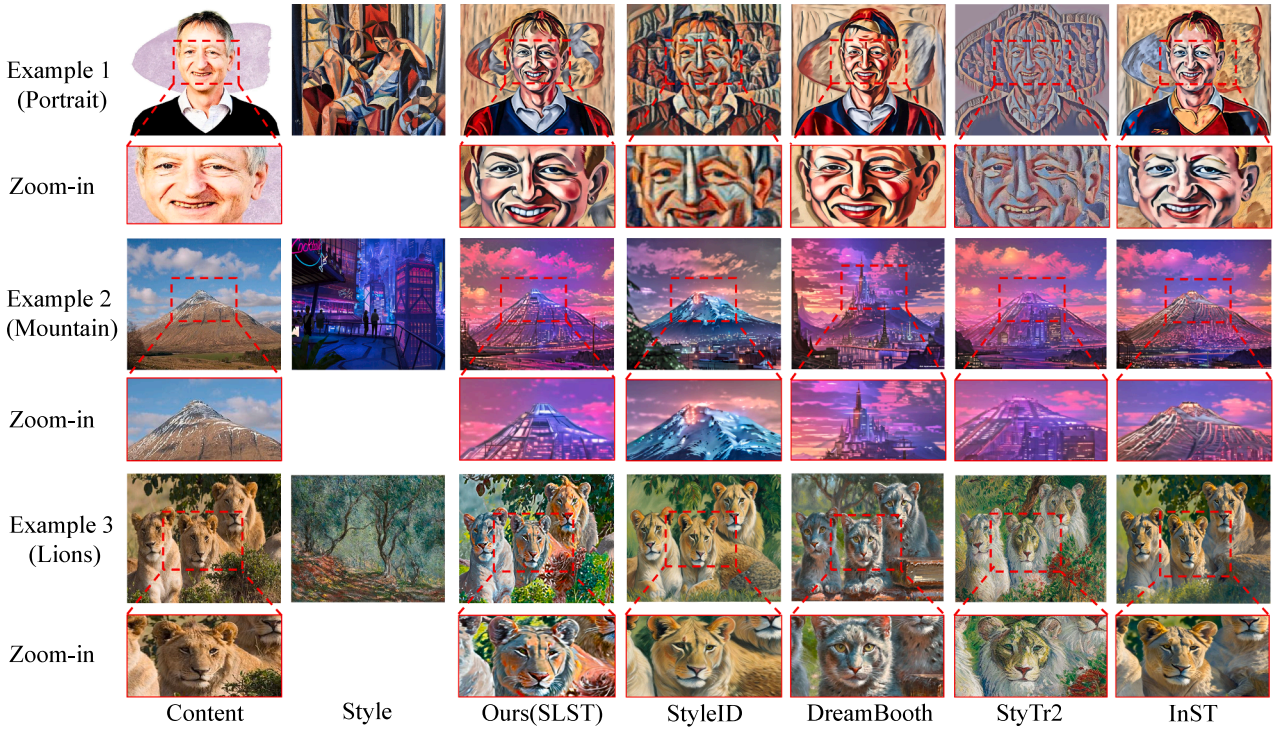


Fig. 4. Visual comparison of SLST and several state-of-the-art methods using paired style and content images. The examples illustrate differences in stylistic consistency and content preservation across the compared methods.

work. Images are resized to $512 \times 512 \times 3$. The random mask blocks are 72×72 pixels, do not overlap, and cover approximately 40% of the image pixels, which are set to $(0, 0, 0)$. The ResBlock and Transformer Encoder are initialized from the pretrained Style-Aware Encoder parameters. Training uses two NVIDIA A40 GPUs with 48 GB memory each and a batch size of 16 per GPU. We use the AdamW optimizer with a learning rate of 1×10^{-4} for 6000 training iterations.

Inference. For inference, we generate a 512×512 image on one NVIDIA A40 (48 GB) GPU using FP16 precision, a batch size of 1, and 50 DDIM denoising steps. The diffusion-generation stage takes approximately 3.4 s per image and uses approximately 17 GB of peak GPU memory. This latency is measured from prepared conditioning inputs and excludes BLIP-2 caption generation and Canny edge extraction; it should therefore be interpreted as denoising-stage latency rather than end-to-end pipeline latency.

Dataset. The training data are collected from LAION-Esthetics (Schuhmann et al., 2022), WikiArt (Phillips & Mackintosh, 2011), and JourneyDB (Sun et al., 2024). Following the style categorization protocol used in StyleShot, we construct a categorized training subset containing 17 major style categories, with 400 images sampled for each category. In addition, 10,000 unclassified artistic images from WikiArt are included to increase stylistic diversity. The implementation is not publicly released at the current submission stage.

For evaluation, we use 24 style types to cover a broad range of artistic appearances. Ten style-reference images are selected for each style type, resulting in 240 style references. Each reference is paired with two content images, producing 480 content-style evaluation pairs. The same evaluation pairs are used for all compared methods.

We distinguish training and evaluation data at the image level. Before sampling, corrupted image files are discarded, and exact duplicate files are identified using SHA-256 hashes. Near duplicates are screened

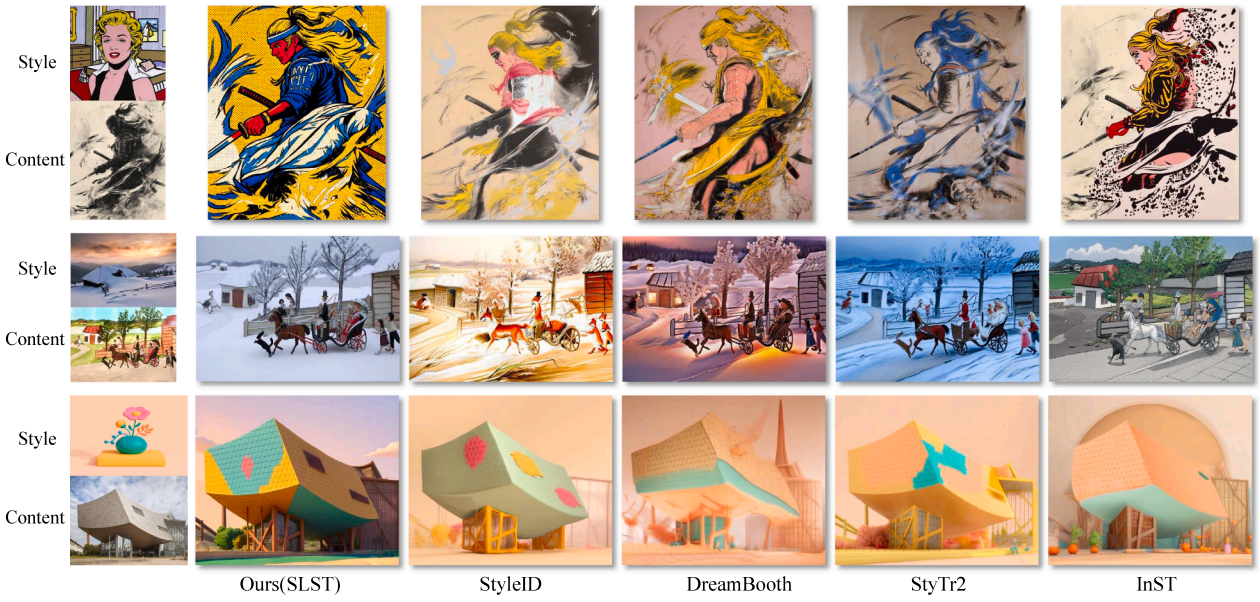


Fig. 5. Visual comparison of SLST and several state-of-the-art methods on highly abstract content images. The examples illustrate differences in stylistic consistency and preservation of the original content structure.

using 64-bit perceptual hashes computed on RGB images resized to a common resolution. Candidate pairs with a Hamming distance of 6 or less are manually inspected, and confirmed duplicate or near-duplicate copies are excluded from the training pool. The training pool is checked against all evaluation style-reference and content images from LAION-Esthetics, WikiArt, and JourneyDB. Any duplicate within the evaluation pool is replaced by a distinct image from the same style group. A fixed random seed of 42 is used for stratified image sampling, and generation seed 1234 is shared across methods for each content-style pair. The same 480 evaluation pairs are used for all methods. The split is image-level rather than style-category-level; therefore, we do not claim strict zero-shot generalization to unseen style categories.

Evaluation Metrics. We use CLIP-S and FID (Heusel et al., 2017) as indicators for evaluating the style similarity of generated images in our experiments. SSIM (Wang et al., 2004) and CFSD (Chung et al., 2024) are used to evaluate the content preservation of generated images relative to content images, and to evaluate the esthetic score and authenticity of generated images using Esthetic[1] and ImageReward (Xu et al., 2023). While no single metric can fully capture the quality of style transfer, we use multiple sets of evaluation metrics so that no single metric dominates the conclusions. Their known limitations can be mitigated through perceptual and esthetic metrics as well as human user evaluation.

- CLIP-S: The feature cosine similarity of the style image and the generated image is calculated to represent the style similarity.
- FID: FID is an indicator used to evaluate the quality of the generated samples of generative model and is often used to measure the similarity of the generated image in distribution with the real image. It evaluates the style fidelity and diversity of the generated image by calculating the distance between two multidimensional Gaussian distributions.
- SSIM: We compute SSIM between the content image and the generated image to measure structural content preservation.
- CFSD: Content feature structure distance (CFSD) is a distance measure that only takes into account spatial correlations between image patches.
- ImageReward: This is a generative image quality evaluation model based on human preference feedback.
- Esthetic: This refers to CLIP + MLP training of the image esthetics score, which evaluates the artistic and esthetic value of images.

4.2. Experiment results

Quantitative results. We compare SLST with existing methods in terms of style similarity, content preservation, and generated-image quality. AdaIN (Huang & Belongie, 2017) is a CNN-based method, StyTr^2 (Deng et al., 2022) is a transformer-based method, and DreamBooth (Ruiz et al., 2023), InST (Zhang et al., 2023b), and StyleID (Chung et al., 2024) use diffusion models. All methods are evaluated on the same input pairs using their official model parameters. Table 1 reports the aggregate results for the evaluated setting. We also replace the SLS-Encoder with a pretrained CLIP image encoder to examine the downstream effect of the style representation. SLST obtains a CLIP-S score of 0.590, an SSIM score of 0.543, and a CFSD score of 0.228 in the reported setting. These aggregate results indicate a favorable observed balance between style similarity and content preservation among the evaluated methods. Because the automatic metrics are reported from a single experimental setting without repeated-run inferential testing, the differences are interpreted descriptively.

User study. We recruited 24 evaluators with different educational backgrounds. Each evaluator viewed three content-style examples, with outputs from all 12 methods arranged in method columns. The order of the 12 method columns was randomized independently for each participant using a pseudorandom generator initialized once with seed 2026, with a new permutation generated for each participant, while the same column order was maintained across the three examples for that participant. After reviewing the examples jointly, each participant selected the two method columns with the highest perceived style similarity and the two with the strongest content preservation. Thus, each participant contributed two selections per criterion, giving 48 selections per criterion across the 24 participants. User-SS and User-CS report the numbers of participants selecting each method for style similarity and content preservation, respectively. The study uses a top-two preference protocol and is not an exhaustive pairwise-comparison experiment.

[1] https://github.com/christophschuhmann/improved-aesthetic-predictor

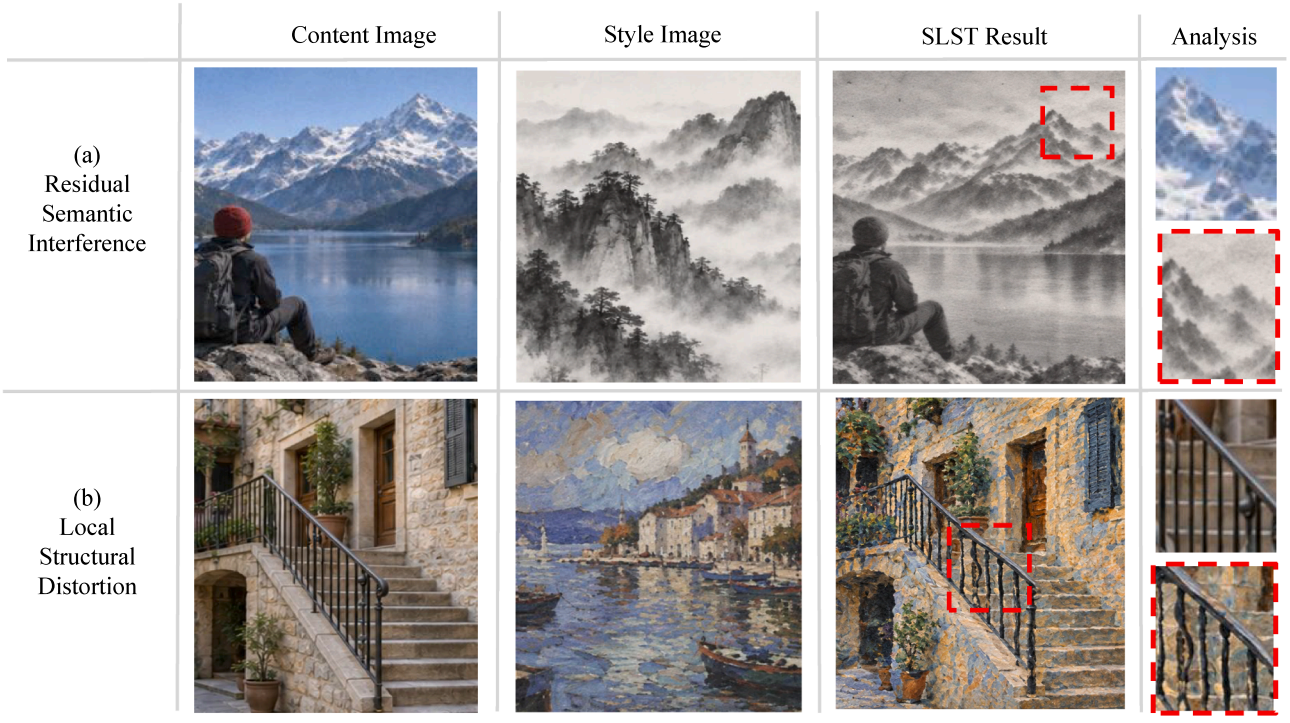


Fig. 6. Representative failure cases of SLST under challenging conditions. (a) Residual semantic interference. Although the overall composition and major content structure are well preserved, a faint reference-specific secondary ridge appears near the main mountain peak. (b) Local structural distortion. The overall layout and stylization remain visually convincing, while fine-grained structures such as the railing may exhibit mild geometric deformation under strong style textures.

The resulting vote counts are presented descriptively in Table 1. Because the table reports aggregate selections and the previously reported inferential analysis was not sufficiently specified to be independently reproduced from the documented results, we do not report Wilcoxon p-values or bootstrap confidence intervals in the revised manuscript. Instead, we report the selection counts and normalized selection rates (the number of participants selecting a method divided by 24). The finite participant pool and subjective nature of the task remain limitations.

Visual comparison. Fig. 4 compares SLST with existing methods across diverse content-style pairs. The style references include abstract and representational works, while the content images include people, landscapes, animals, and other scenes. The examples illustrate differences in color transfer, local texture, and preservation of object structure across methods. In the displayed cases, SLST retains the major content layout while transferring appearance characteristics from the reference style; residual semantic interference and local structural distortion may still occur in challenging cases. StyleID (Chung et al., 2024) aligns text features with style features to avoid interference between style control and text control, but it alters the generated results and causes color distortion. DreamBooth (Ruiz et al., 2023) requires multiple reference images for model fine-tuning, and when only one style reference image and one content image are available, it leads to changes in object features and the appearance of unrelated items. StyTr^2 (Deng et al., 2022) can accurately capture the tone and texture features of the reference image, but it performs weakly on complex styles, lacks consistency in the details of generated results, and causes content leakage from the style image into the final generated image. InST (Zhang et al., 2023b) employs an attention-based text inversion method to invert style embeddings into text embeddings, yet it fails to effectively transfer the color application from the style image to the content image when there is a significant discrepancy between the two. Our SLST method, by contrast, learns fine-grained style-dominant representations with reduced content interference and achieves a favorable empirical balance between style fidelity and content preservation in the evaluated cases.

We also perform visual experiments in complex content scenarios, and the results are shown in Fig. 1. The first line uses complex architectural scenes, while the second line uses dense crowd scenes. All methods have good visual performance in the color of style transfer, but there is a big gap in content preservation in complex scenarios. As shown in the red box in Fig. 1, neither InST (Zhang et al., 2023b) nor StyTr^2 (Deng et al., 2022) can maintain the shape of the dark clouds in the original content image in the first scene. Although StyleID (Chung et al., 2024) and DreamBooth (Ruiz et al., 2023) maintain the shape of the clouds, the color of the clouds is assimilated with the surrounding buildings and trees. In the red box of the second scene, the luggage box generated by StyleID (Chung et al., 2024) is blurred and the shape is distorted. The number of luggage boxes generated by InST (Zhang et al., 2023b) is inconsistent. DreamBooth (Ruiz et al., 2023) does not generate a suitcase, and the luggage box becomes other items in the result of StyTr^2 (Deng et al., 2022) generation.

In the evaluated examples, several methods show difficulties with highly abstract or non-representational content images. In particular, InST (Zhang et al., 2023b) and StyleID (Chung et al., 2024) may alter abstract structures or semantic layouts during style transfer. Such changes can include structural distortion or loss of salient features. In the examples shown in Fig. 5, SLST appears to preserve some key structural and semantic elements while transferring the reference style. These qualitative observations suggest that abstract content can be challenging for current style-transfer methods, but do not constitute a direct measurement of content information in the learned style representation.

Failure-case analysis. To provide a more balanced evaluation of the proposed method, we analyze representative cases in which SLST

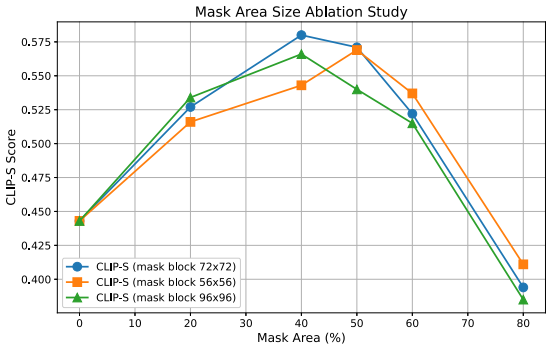


Fig. 7. Ablation of the mask area size. The horizontal axis is the percentage of the total mask area to the style image area. The point at 0% denotes the untrained initialization baseline and is not a trained model with zero masking. The total area of the image is 512×512, and the mask block sizes of the three groups of experiments are 56×56 , 72×72 and 96×96 respectively.

does not fully preserve the desired balance between style transfer and content structure, as shown in Fig. 6. Although SLST generally achieves favorable style fidelity and content preservation, two recurring limitations can still be observed in challenging cases.

First, when the style reference contains strong structural or semantic patterns, a small amount of reference-specific information may remain in the generated image. As illustrated in Fig. 6(a), the primary mountain structure and overall scene layout are preserved, while a faint secondary ridge that is not present in the original content image appears near the main peak.

Second, strong style textures may interfere with fine-grained geometric structures. In Fig. 6(b), the overall architectural layout and stylization remain consistent with the input, but several thin railing elements exhibit slight deformation after stylization.

These cases indicate that the proposed masking-based style representation reduces, rather than completely eliminates, residual semantic interference, and that a trade-off may still exist between strong stylization and fine-grained structural preservation.

5. Ablation study

5.1. Verification of SLS-Encoder

The behavior of the SLS-Encoder is influenced by the proportion of the image masked during training. We vary the mask-area ratio and examine its association with convergence and downstream style-transfer results. In the reported experiments, a mask ratio of 20% may leave more visible semantic content, whereas a ratio around 40% provides the best observed balance among the tested settings between disrupting content cues and retaining stylistic details. Larger mask ratios may remove useful style information and hinder convergence. The size of individual mask blocks also matters: small blocks may leave important spatial relationships visible, while large blocks may remove local stylistic details. Under the tested settings, a block size of 72×72 pixels yields the most favorable observed balance. These findings describe the evaluated configurations and do not establish a universally optimal masking strategy (see Fig. 7).



Fig. 8. Visual comparison of feature concatenation. The examples show differences in color distribution and boundary transitions when style and edge embeddings are supplied separately or concatenated.

To evaluate whether the SLS-Encoder can be transferred to other architectures, we integrate it into StyleID (Chung et al., 2024) and InST (Zhang et al., 2023b), replacing their original style encoders while keeping the remaining components unchanged. The results are reported in Table 2. The reported changes provide downstream evidence about how SLS-Encoder affects style-transfer behavior in these two frameworks. They do not directly measure semantic information retained in the embeddings or establish representation-level disentanglement.

5.2. Verification of SLST architecture

Fig. 8 compares outputs generated with and without concatenating the style and edge embeddings. When the two embeddings are supplied separately, the illustrated results show more saturated colors and less natural boundary transitions. With concatenation, the displayed examples show more coordinated distant peaks and a clearer water reflection. This qualitative comparison suggests that feature concatenation can help combine style and structural cues in the evaluated example.

To examine the contribution of text embeddings, we remove the text condition and rely on edge images for content preservation. As shown in Table 3, SSIM decreases from 0.543 to 0.505 (approximately 7.00%), while CFSD increases from 0.228 to 0.242 (approximately 6.14%). This result suggests that text guidance contributes to content preservation in the evaluated setting, as edge information alone may not fully represent complex semantic content.

5.3. Component-level ablation

To analyze the contribution of each design choice, we ablate key components of SLST and report the aggregate results in Table 4. Removing the variance-alignment term changes CLIP-S from 0.590 to 0.581, SSIM from 0.543 to 0.535, and CFSD from 0.228 to 0.233. Removing the mean-alignment term yields CLIP-S 0.580, SSIM 0.532, and CFSD 0.237. These changes suggest that both alignment terms contribute to the observed results under the evaluated setting. Replacing the separate text/style cross-attention branches with shared cross-attention yields the largest observed decline among the component ablations: CLIP-S changes from 0.590 to 0.572, SSIM from 0.543 to 0.525, and CFSD from 0.228 to 0.243. This pattern suggests that separate conditioning paths contribute to the measured performance in the evaluated setting; it does not establish statistical significance or prove that this component is universally dominant.

Table 2
Cross-architecture evaluation of SLS-Encoder. The table reports downstream style-similarity and content-preservation metrics when SLS-Encoder replaces the original style encoder in StyleID (Chung et al., 2024) and InST (Zhang et al., 2023b). Values in parentheses indicate relative changes compared with the original implementations.

Method	CLIP-S ↑	SSIM ↑	FID ↓	CFSD ↓
StyleID (orig)	0.569	0.398	18.831	0.251
StyleID + SLS-Enc	0.587 (+3.2%)	0.415 (+4.3%)	18.500 (-1.8%)	0.243 (-3.2%)
InST (orig)	0.538	0.507	21.311	0.676
InST + SLS-Enc	0.565 (+5.0%)	0.518 (+2.2%)	20.100 (-5.7%)	0.662 (-2.1%)

Table 3
Results of the ablation study on text embedding in the Self-Learning Style Transfer architecture. SLST (Full) is the evaluation result of adding text embedding, and SLST (No Text) is the result of not adding text embedding.

	CLIP-S↑	SSIM↑	CFSD↓	Aesth↑
SLST (Full)	**0.590**	**0.543**	**0.228**	**5.411**
SLST (No Text)	0.577	0.505	0.242	5.266

Table 4
Component ablation of the proposed SLST framework. The second and third rows report results without variance alignment and mean alignment, respectively. The fourth row replaces decoupled cross-attention with shared cross-attention. Values are aggregate results from the reported setting and are interpreted descriptively.

Method	CLIP-S↑	SSIM↑	CFSD↓	Aesth↑
SLST (full)	0.590	0.543	0.228	5.411
w/o variance alignment	0.581	0.535	0.233	5.400
w/o mean alignment	0.580	0.532	0.237	5.389
w/o decoupled cross-attention	0.572	0.525	0.243	5.374

6. Conclusion

In this work, we propose Self-Learning Style Transfer (SLST), a diffusion-based framework designed to reduce style-content interference while maintaining stylistic fidelity and content structure. The SLS-Encoder uses random masking and cross-view consistency learning to encourage style-dominant representations with reduced content dependence. The reported quantitative, qualitative, and user-study results indicate a favorable balance between style similarity and content preservation in the evaluated settings. Residual semantic interference and local structural distortion may still occur under challenging conditions, and the current results do not establish complete representation-level disentanglement.

7. Limitations and future work

Although SLST achieves a favorable balance between style fidelity and content preservation, several limitations remain. First, as demonstrated by the failure cases in Fig. 6, highly salient structural or semantic patterns in the style reference may occasionally introduce weak residual interference into the generated image. Second, strong style textures may affect fine-grained geometric structures, such as thin railings or repeated local details, even when the global layout is well preserved. Third, the current method controls style at the global image level and does not yet support explicit region-specific style manipulation. Finally, our evaluation mainly focuses on downstream generation behavior, and direct representation-level analyses, such as embedding probing or explicit content-leakage classification, remain future work. The automatic metrics are reported as aggregate results under the evaluated settings without repeated-run inferential testing. The user study also relies on a finite participant pool and a subjective preference-selection protocol.

CRediT authorship contribution statement

Jing Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology; **Ying Li:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Conceptualization; **Shuaiyu Deng:** Visualization; **Longye Du:** Validation; **Yanjun Chen:** Writing – review & editing; **Jun Guo:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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